THE FREE HEYTING CATEGORY ON A PRETOPOS IS A PRETOPOS ONLY IN THE BOOLEAN CASE: THE INTUITIONISTIC TRANSLATION AND THE PRETOPOS COMPLETION

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Abstract. Reading a coherent theory in intuitionistic first-order logic is the free Heyting category $\mathsf{H}\mathcal{C}$ on its coherent syntactic category $\mathcal{C}$; adding the imaginaries is the pretopos completion P. We prove that the two operations commute only trivially: the canonical functor $\mathsf{HP}\mathcal{C} \to \mathsf{PH}\mathcal{C}$ is always full and faithful, and it is an equivalence if and only if $\mathcal{C}$ is Boolean, in which case H does nothing. In particular the free Heyting category on a pretopos is a pretopos only when the pretopos is Boolean. The objects we show to be missing from $\mathsf{HP}\mathcal{C}$ are two copies of an object glued along the intuitionistic negation of a non-complemented subobject; they are excluded by an invariant of the objects of $\mathsf{HP}\mathcal{C}$ in Kripke models over a fork, whose existence for every non-Boolean $\mathcal{C}$ is a form of the homomorphism preservation theorem. H commutes with the positive completion; with the exact completion it commutes for propositional theories and fails when some object has two points in the source model of a homomorphism crossing a subobject. Since the Morleyization is H followed by the addition of excluded middle, which commutes with P, the commutation of the Morleyization with P, governed by a Galois criterion, is the statement that excluded middle repairs a comparison which, for non-Boolean $\mathcal{C}$, is not an equivalence before it.

1. Introduction

A coherent theory T has a coherent syntactic category $\mathcal{C}_T$, and three constructions on it are the subject of this paper and of its predecessor [3]. The pretopos completion $\mathsf{P}\mathcal{C}$ adds finite coproducts of sorts and quotients by definable equivalence relations, the coherent imaginaries [1, A3.3, D1.4]. The Morleyization $\mathsf{B}\mathcal{C}$ is the syntactic category of T read in classical first-order logic. Between them lies the Heyting completion $\mathsf{H}\mathcal{C}$, the syntactic category of T read in intuitionistic first-order logic, and $\mathsf{B} \simeq \mathsf{KH}$ with K the addition of excluded middle [3, §9].

2020 *Mathematics Subject Classification.* 03G30 (primary), 18F10, 03F55, 03C90 (secondary).

Key words and phrases. Heyting category, pretopos completion, coherent category, intuitionistic first-order logic, Kripke model, imaginaries, Morleyization, Boolean category.

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In [3] the order of translation and completion was studied for B: there is a full and faithful comparison $\mathsf{BP}\mathcal{C} \to \mathsf{PB}\mathcal{C}$, which is an equivalence exactly when a Galois criterion holds [3, Thm. 6.1]; it fails for the theory of complete multipartite graphs and holds, for example, for the theory of an equivalence relation. It was also shown that K commutes with P on all Heyting categories [3, Thm. 9.2], so that the comparison for B is the image under K of the comparison

$$c_{\mathcal{C}} \colon \mathsf{HP}\mathcal{C} \longrightarrow \mathsf{PH}\mathcal{C}$$

for H. The question left open was when $c_{\mathcal{C}}$ itself is an equivalence. The answer is the following.

Theorem 1.1. *For a small coherent category $\mathcal{C}$ the functor $c_{\mathcal{C}}$ is full and faithful, and the following are equivalent.*

(i) *$c_{\mathcal{C}} \colon \mathsf{HP}\mathcal{C} \to \mathsf{PH}\mathcal{C}$ is an equivalence.*
(ii) *$\mathsf{HP}\mathcal{C}$ is a pretopos.*
(iii) *$\mathcal{C}$ is Boolean.*

If they hold, the units $\mathcal{C} \to \mathsf{H}\mathcal{C}$ and $\mathsf{P}\mathcal{C} \to \mathsf{HP}\mathcal{C}$ are equivalences.

For a pretopos $\mathcal{C}$ this says that the free Heyting category on $\mathcal{C}$ is a pretopos if and only if $\mathcal{C}$ is Boolean (Corollary 5.1).

What is new and what is known. New, to the author's knowledge: Theorem 1.1; the invariant that proves it (Lemma 3.1); the commutation of H with the positive completion (Proposition 6.2) and the resulting location of the failure (Corollary 6.3, Proposition 6.4); and the reading of the Galois criterion of [3] as the statement that $c_{\mathcal{C}}$ is inverted by K (Corollary 7.1). Known and used: syntactic categories of first-order theories are Heyting categories and classify their models in Heyting categories [1, D1.4]; the pretopos completion [1, A3.3]; completeness of coherent logic for set-valued models [1, D1.5], [2]; the homomorphism preservation theorem of Łoś, Tarski and Lyndon, of which Lemma 4.1 is the form needed here and is proved in full; the Heyting structure of presheaf categories, that is, Kripke semantics; and from [3] the facts that $\mathsf{P}\mathcal{D}$ is Heyting when $\mathcal{D}$ is, with the expected universal property, and that K commutes with P.

Conventions. Coh is the $(2,1)$-category of small coherent categories, coherent functors and natural isomorphisms, and **Heyt** that of small Heyting categories (coherent categories in which every pullback functor $f^* \colon \mathrm{Sub}(B) \to \mathrm{Sub}(A)$ has a right adjoint $\forall_f$), Heyting functors (coherent functors preserving the $\forall_f$) and natural isomorphisms [1, A1.4]. A Heyting functor preserves the implication of subobjects and the pseudo-complement $\neg U = (U \Rightarrow 0)$, since $U \Rightarrow V = \forall_i(i^* V)$ for the inclusion $i \colon U \to A$. A coherent category is *Boolean* if every subobject is complemented. A model of $\mathcal{C}$ is a coherent functor $\mathcal{C} \to \mathbf{Set}$, and $\mathrm{Mod}(\mathcal{C})$ is the category of models with all natural transformations, the homomorphisms. The universal properties

below are stated for small categories but hold, with the same proofs, for targets such as $\mathbf{Set}^{\mathcal{K}}$, $\mathcal{K}$ a small category; alternatively one replaces $\mathbf{Set}^{\mathcal{K}}$ by a small full subcategory closed under the operations of a Heyting pretopos and containing the objects in question.

2. The two completions and the comparison

The Heyting completion. For $\mathcal{C}$ in **Coh** let $\mathsf{H}\mathcal{C}$ be the syntactic category of the internal coherent theory of $\mathcal{C}$ (a sort for every object, a function symbol for every morphism, the coherent sequents valid in $\mathcal{C}$) read in intuitionistic first-order logic with possibly empty sorts (the initial object is a sort), and $\eta\colon \mathcal{C} \to \mathsf{H}\mathcal{C}$ the coherent functor sending an object to its sort. $\mathsf{H}\mathcal{C}$ is a Heyting category, and for every Heyting category $\mathcal{D}$ restriction along η is an equivalence of groupoids

$$\mathbf{Heyt}(\mathsf{H}\mathcal{C}, \mathcal{D}) \;\simeq\; \mathbf{Coh}(\mathcal{C}, \mathcal{D})\,:$$

a coherent functor $\mathcal{C} \to \mathcal{D}$ is a model of the internal theory in $\mathcal{D}$, it interprets every first-order formula, intuitionistic logic is sound in Heyting categories, and an isomorphism of models preserves the interpretations of all formulas [1, D1.4], [3, §9]. Every object of $\mathsf{H}\mathcal{C}$ is a subobject of some ηA: it is a formula in a context $x_1{:}A_1, \ldots, x_n{:}A_n$, a subobject of $\eta(A_1 \times \cdots \times A_n)$.

The forgetful 2-functor $\mathbf{Heyt} \to \mathbf{Coh}$ is not full, a coherent functor between Heyting categories need not be Heyting. So H is a free construction and not a reflector; the join lemma of [4, 3], which concerns reflectors, does not apply to it, and we compare the composites HP and PH directly.

Lemma 2.1. *If $\mathcal{C}$ is Boolean, then $\eta\colon \mathcal{C} \to \mathsf{H}\mathcal{C}$ is an equivalence.*

Proof. A Boolean coherent category is Heyting, with $\forall_f U = \neg\exists_f \neg U$. Let $G\colon \mathcal{C} \to \mathcal{D}$ be coherent with $\mathcal{D}$ Heyting. Then G preserves complements and images, so $G(\forall_f U)$ is a complement W of $\exists_{Gf} V'$, where $V' = G(\neg U)$ is a complement of $V = GU$. In a Heyting category this W is $\forall_{Gf} V$: for a subobject W', $(Gf)^* W' \le V$ iff $(Gf)^* W' \wedge V' = 0$ iff $W' \wedge \exists_{Gf} V' = 0$ (Frobenius reciprocity) iff $W' \le W$. So every coherent functor from $\mathcal{C}$ to a Heyting category is Heyting, and the identity of $\mathcal{C}$ has the universal property of η. □

The pretopos completion of a Heyting category. We write $\iota\colon \mathcal{C} \to \mathsf{P}\mathcal{C}$ for the unit of the pretopos completion; it is full and faithful, and restriction along it is an equivalence between the categories of coherent functors $\mathsf{P}\mathcal{C} \to \mathcal{E}$ and $\mathcal{C} \to \mathcal{E}$ with all natural transformations (not only the groupoids $\mathbf{Coh}(\mathsf{P}\mathcal{C}, \mathcal{E})$ and $\mathbf{Coh}(\mathcal{C}, \mathcal{E})$), when $\mathcal{E}$ is a pretopos [1, A3.3]. In particular $\mathrm{Mod}(\mathsf{P}\mathcal{C}) \simeq \mathrm{Mod}(\mathcal{C})$. Every object of $\mathsf{P}\mathcal{C}$ is the quotient of a finite coproduct of objects ιA by an equivalence relation.

Lemma 2.2 ([3, Lemma 9.1])**.** *If $\mathcal{D}$ is a Heyting category, then $\mathsf{P}\mathcal{D}$ is a Heyting pretopos and $\iota\colon \mathcal{D} \to \mathsf{P}\mathcal{D}$ is a Heyting functor; and for a Heyting*

pretopos $\mathcal{E}$ a coherent functor $\mathsf{P}\mathcal{D} \to \mathcal{E}$ is Heyting if and only if its restriction to $\mathcal{D}$ is.

Consequently $\mathsf{PH}\mathcal{C}$ is the free Heyting pretopos on $\mathcal{C}$: $\mathbf{Heyt}(\mathsf{PH}\mathcal{C}, \mathcal{E}) \simeq \mathbf{Coh}(\mathcal{C}, \mathcal{E})$ for Heyting pretoposes $\mathcal{E}$.

The comparison. The coherent functor $\mathsf{P}\eta\colon \mathsf{P}\mathcal{C} \to \mathsf{PH}\mathcal{C}$ has a Heyting target, so it extends, uniquely up to a unique isomorphism, to a Heyting functor

$$\begin{array}{ccc} \mathsf{P}\mathcal{C} & \xrightarrow{\eta_{\mathsf{P}\mathcal{C}}} & \mathsf{HP}\mathcal{C} \\ & \searrow^{\mathsf{P}\eta} & \downarrow{\scriptstyle c_{\mathcal{C}}} \\ & & \mathsf{PH}\mathcal{C}. \end{array}$$

Lemma 2.3. *$c_{\mathcal{C}}$ is full and faithful. It is an equivalence if and only if $\mathsf{HP}\mathcal{C}$ is a pretopos.*

Proof. For a Heyting pretopos $\mathcal{E}$, by Lemma 2.2 and the universal properties,

$$\begin{aligned} \mathbf{Heyt}(\mathsf{PHP}\mathcal{C}, \mathcal{E}) &\simeq \mathbf{Heyt}(\mathsf{HP}\mathcal{C}, \mathcal{E}) \simeq \mathbf{Coh}(\mathsf{P}\mathcal{C}, \mathcal{E}) \\ &\simeq \mathbf{Coh}(\mathcal{C}, \mathcal{E}) \simeq \mathbf{Heyt}(\mathsf{PH}\mathcal{C}, \mathcal{E}), \end{aligned}$$

by restriction to $\mathcal{C}$. So there is a Heyting equivalence $e\colon \mathsf{PHP}\mathcal{C} \simeq \mathsf{PH}\mathcal{C}$ compatible with the functors from $\mathcal{C}$, and $e \circ \iota_{\mathsf{HP}\mathcal{C}}$ is a Heyting functor $\mathsf{HP}\mathcal{C} \to \mathsf{PH}\mathcal{C}$ whose restriction to $\mathsf{P}\mathcal{C}$ is a coherent functor into a pretopos restricting on $\mathcal{C}$ as $\mathsf{P}\eta$ does; hence $e \circ \iota_{\mathsf{HP}\mathcal{C}} \cong c_{\mathcal{C}}$. The unit $\iota_{\mathsf{HP}\mathcal{C}}$ is full and faithful, and it is an equivalence if and only if $\mathsf{HP}\mathcal{C}$ is a pretopos. □

3. Kripke models and an invariant

Let $\mathcal{K}$ be a small category. The category $\mathbf{Set}^{\mathcal{K}}$ is a Heyting pretopos, and limits, images, unions, coproducts and quotients in it are computed at each object of $\mathcal{K}$; so a coherent functor $\mathcal{C} \to \mathbf{Set}^{\mathcal{K}}$ is the same as a functor $M\colon \mathcal{K} \to \mathrm{Mod}(\mathcal{C})$, a *Kripke model* of $\mathcal{C}$ over $\mathcal{K}$; the transitions are arbitrary homomorphisms and need not be injective. We write M_k for the model at k. The pseudo-complement in $\mathbf{Set}^{\mathcal{K}}$ is the Kripke negation: for a subfunctor $S \subseteq A$,

$$(\neg S)(k) = \{x \in A(k) \mid A(w)x \notin S(l) \text{ for every arrow } w\colon k \to l\}, \tag{1}$$

since the right-hand side is a subfunctor disjoint from S, and every subfunctor W disjoint from S lies in it ($x \in W(k)$ gives $A(w)x \in W(l)$).

A Kripke model M extends to

$$\begin{aligned} M^{\mathsf{H}}&\colon \mathsf{H}\mathcal{C} \to \mathbf{Set}^{\mathcal{K}}, & M^{\mathsf{P}}&\colon \mathsf{P}\mathcal{C} \to \mathbf{Set}^{\mathcal{K}}, \\ M^{\mathsf{HP}}&\colon \mathsf{HP}\mathcal{C} \to \mathbf{Set}^{\mathcal{K}}, & M^{\mathsf{PH}}&\colon \mathsf{PH}\mathcal{C} \to \mathbf{Set}^{\mathcal{K}}, \end{aligned}$$

the first, third and fourth Heyting (Lemma 2.2), with $M^{\mathsf{HP}}\eta_{\mathsf{P}\mathcal{C}} \cong M^{\mathsf{P}}$ and

$$M^{\mathsf{PH}} \circ c_{\mathcal{C}} \;\cong\; M^{\mathsf{HP}}, \tag{2}$$

both sides being Heyting functors whose restrictions to $\mathsf{P}\mathcal{C}$ are coherent functors into a pretopos restricting to M on $\mathcal{C}$. The functor M^{P} is the Kripke model of $\mathsf{P}\mathcal{C}$ corresponding to M under $\mathrm{Mod}(\mathsf{P}\mathcal{C}) \simeq \mathrm{Mod}(\mathcal{C})$: for Y in $\mathsf{P}\mathcal{C}$, the set $M^{\mathsf{P}}(Y)(k)$ is the value at Y of the extension of M_k, and an arrow w of $\mathcal{K}$ acts by the extension of the homomorphism $M(w)$.

Lemma 3.1 (invariant)**.** *Let X be an object of $\mathsf{HP}\mathcal{C}$, M a Kripke model of $\mathcal{C}$ over $\mathcal{K}$, and $w\colon k \to l$ an arrow of $\mathcal{K}$ such that $M(w)$ is an isomorphism of models. Then $M^{\mathsf{HP}}(X)(w)$ is injective.*

Proof. X is a subobject of $\eta_{\mathsf{P}\mathcal{C}}Y$ for an object Y of $\mathsf{P}\mathcal{C}$, so $M^{\mathsf{HP}}(X)$ is a subfunctor of $M^{\mathsf{P}}(Y)$. The functor $\mathcal{K} \to \mathrm{Mod}(\mathsf{P}\mathcal{C})$ corresponding to M^{P} sends w to an isomorphism, since $M(w)$ is one and $\mathrm{Mod}(\mathsf{P}\mathcal{C}) \simeq \mathrm{Mod}(\mathcal{C})$; so $M^{\mathsf{P}}(Y)(w)$ is a bijection and its restriction is injective. □

When $\mathcal{C}$ is not Boolean, some objects of $\mathsf{PH}\mathcal{C}$ do not have this property (Section 5): they are quotients by relations whose truth at k depends on all the arrows out of k and not only on the model M_k.

4. Non-Boolean categories have crossing homomorphisms

Lemma 4.1. *Let $S \le A$ be a subobject in a small coherent category $\mathcal{C}$ which is not complemented. Then there are models M, N of $\mathcal{C}$, a homomorphism $h\colon M \to N$ and $a \in A(M) \setminus S(M)$ with $h_A(a) \in S(N)$.*

Proof. For $U \le B$ with inclusion i we write $U(x)$ for the formula $\exists u{:}U\,(i(u) = x)$. We use the completeness theorem: if $U, U_1, \dots, U_n \le B$ and $U(M) \subseteq U_1(M) \cup \dots \cup U_n(M)$ for all models M, then $U \le U_1 \vee \dots \vee U_n$ [1, D1.5], [2]; and the compactness theorem for the classical first-order theory $T_{\mathcal{C}}$ whose models are the models of $\mathcal{C}$, in many-sorted logic with possibly empty sorts (the initial object is a sort).

Let $I = \{U \le A \mid U \wedge S = 0\}$, an ideal of the distributive lattice $\mathrm{Sub}(A)$, and J the ideal of the W with $W \le S \vee U$ for some $U \in I$. It is proper: $A \le S \vee U$ with $U \wedge S = 0$ would make U a complement of S. By the prime filter theorem there is a prime filter $\mathfrak{p}$ of $\mathrm{Sub}(A)$ disjoint from J.

There are a model M and $a \in A(M)$ with $a \in U(M)$ iff $U \in \mathfrak{p}$, for all $U \le A$: the theory $T_{\mathcal{C}}$ with a new constant c of sort A and the axioms $U(c)$, $U \in \mathfrak{p}$, and $\neg U(c)$, $U \notin \mathfrak{p}$, is finitely satisfiable, since otherwise there are $U_0 \in \mathfrak{p}$ and $U_1, \dots, U_n \notin \mathfrak{p}$ with $U_0(M) \subseteq \bigcup_i U_i(M)$ in all models, hence $U_0 \le \bigvee_i U_i$, and $\bigvee_i U_i \in \mathfrak{p}$ contradicts the primeness of $\mathfrak{p}$ (for $n = 0$, $0 \in \mathfrak{p}$ contradicts its properness). In particular $a \notin S(M)$, and $a \notin U(M)$ for $U \in I$.

Now add to $T_{\mathcal{C}}$ a constant c_b of sort B for every object B and every $b \in B(M)$, the axioms

$$f(c_b) = c_{f_M(b)} \qquad (f\colon B \to B' \text{ a morphism of } \mathcal{C},\ b \in B(M)),$$

and the axiom $S(c_a)$. A finite part of these axioms mentions, besides c_a, finitely many distinct constants $c_{b_1}, \dots, c_{b_n}$, $b_i \in B_i(M)$, and finitely many

equations between them and c_a. Put $b_0 = a$, $B_0 = A$ and $B = \prod_{i=0}^{n} B_i$, and let $\theta \leq B$ be the subobject defined by the conjunction of these equations, one variable y_i replacing each constant c_{b_i} (an equation $f(y_i) = y_j$ defines an equaliser, a subobject of B); then $(a, b_1, \ldots, b_n) \in \theta(M)$. If the finite part had no model, every model N' would satisfy $\exists_{\pi_0}\theta(N') \cap S(N') = \emptyset$, so $U := \exists_{\pi_0}\theta \leq A$ would satisfy $U \wedge S = 0$ by completeness, that is $U \in I \subseteq J$; but $a \in U(M)$ gives $U \in \mathfrak{p}$, a contradiction. By compactness there is a model N of all these axioms, and $h_B(b) := c_b^N$ is, by the axioms, a natural transformation $M \to N$, with $h_A(a) \in S(N)$. □

Remark 4.2. Lemma 4.1 is the homomorphism preservation theorem in the form: a subobject whose complement is preserved by all homomorphisms of models is complemented. Its converse is trivial, so a coherent category is Boolean if and only if every homomorphism of its models reflects every subobject. In that case every first-order formula is equivalent to a coherent one and the homomorphisms are the elementary embeddings of the models of the corresponding classical theory. (A single homomorphism reflecting all subobjects need not be elementary: the inclusion of a two-element set in a three-element set, for pure equality.)

5. Proof of the main theorem

Proof of Theorem 1.1. The first assertion and (i)⇔(ii) are Lemma 2.3.

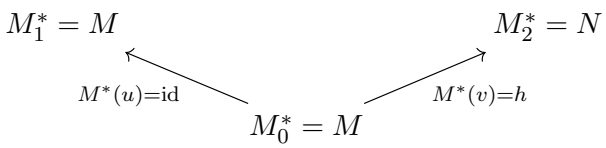
(iii)⇒(i). If $\mathcal{C}$ is Boolean, so is $\mathsf{P}\mathcal{C}$ [3, Prop. 3.4]: complements are taken componentwise in finite coproducts, and the complement of a saturated subobject is saturated. By Lemma 2.1, $\eta_{\mathcal{C}}$ and $\eta_{\mathsf{P}\mathcal{C}}$ are equivalences; so $\mathsf{HP}\mathcal{C} \simeq \mathsf{P}\mathcal{C}$ is a pretopos.

(i)⇒(iii). Suppose that $S \leq A$ is not complemented in $\mathcal{C}$, and let $h\colon M \to N$ and a be as in Lemma 4.1. In $\mathsf{PH}\mathcal{C}$ let $\neg S \leq A$ be the pseudo-complement of (the image of) S in (the image of) A, and let

$$Q \;:=\; (A + A)/E, \qquad E := \Delta_{A+A} \vee (\kappa_1, \kappa_2)(\neg S) \vee (\kappa_2, \kappa_1)(\neg S),$$

with $\kappa_1, \kappa_2\colon A \to A + A$ the coprojections and $(\kappa_i, \kappa_j)(\neg S)$ the image of $\neg S$ in $(A + A) \times (A + A)$: the two copies of A glued along $\neg S$, the pushout of $A \leftarrow \neg S \to A$. The relation E is an equivalence relation, so Q exists in the pretopos $\mathsf{PH}\mathcal{C}$. We show that Q is not isomorphic to $c_{\mathcal{C}}(X)$ for any object X of $\mathsf{HP}\mathcal{C}$.

Let $\mathcal{K}$ be the fork, the poset with three elements $0 < 1$ and $0 < 2$, with arrows $u\colon 0 \to 1$ and $v\colon 0 \to 2$, and M^* the Kripke model



Since $M^{*\mathsf{PH}}$ is a Heyting functor between pretoposes, it preserves pseudo-complements, coproducts, images and quotients; so $M^{*\mathsf{PH}}(Q)$ is the quotient

of $M^*(A) + M^*(A)$ which identifies the two copies of $x \in A(M_k^*)$ exactly when $x \in (\neg M^*(S))(k)$. By (1), $a \in (\neg M^*(S))(1)$, because the only arrow out of 1 is the identity and $a \notin S(M)$; and $a \notin (\neg M^*(S))(0)$, because $M^*(A)(v)a = h_A(a) \in S(N)$. Hence the two copies of a are distinct points of $M^{*\mathsf{PH}}(Q)(0)$ with the same image under u in $M^{*\mathsf{PH}}(Q)(1)$.

If $Q \cong c_{\mathcal{C}}(X)$, then by (2) $M^{*\mathsf{PH}}(Q) \cong M^{*\mathsf{HP}}(X)$, whose map along u is injective by Lemma 3.1, since $M^*(u)$ is an isomorphism. This is a contradiction; so $c_{\mathcal{C}}$ is not essentially surjective.

The last assertion is Lemma 2.1 for $\mathcal{C}$ and $\mathsf{P}\mathcal{C}$. □

Corollary 5.1. *Let $\mathcal{C}$ be a small pretopos. The free Heyting category $\mathsf{H}\mathcal{C}$ on $\mathcal{C}$ is a pretopos if and only if $\mathcal{C}$ is Boolean, and then $\mathsf{H}\mathcal{C} \simeq \mathcal{C}$.*

Proof. $\iota\colon \mathcal{C} \to \mathsf{P}\mathcal{C}$ is an equivalence, so $\mathsf{H}\mathcal{C} \simeq \mathsf{HP}\mathcal{C}$; apply Theorem 1.1. □

Example 5.2 (pure equality)**.** Let $\mathcal{C}$ be the syntactic category of the coherent theory with one sort σ and no symbols and axioms. The diagonal of $\sigma \times \sigma$ is not complemented; a crossing homomorphism is the map from a two-element set $\{a, b\}$ to a one-element set, with the pair (a, b). The object Q is two copies of $\sigma \times \sigma$ glued along $\neg(x = y)$: an intuitionistic imaginary of the theory of pure equality which is not an intuitionistically definable subset of a coherent imaginary. Over the fork, $\neg(x = y)$ is empty at 0 and is the set of the two off-diagonal pairs at 1; so the fibre of Q at 0 has eight points, two copies of the four pairs, and the fibre at 1 has six, the two copies of (a, b) and of (b, a) being identified.

Example 5.3 (propositional theories)**.** Let $\mathcal{C} = D$ be a bounded distributive lattice, a propositional coherent theory. Up to isomorphism its models are the prime filters and the homomorphisms are the inclusions, and Lemma 4.1 is the classical fact that a distributive lattice in which the prime filters form an antichain is Boolean. $\mathsf{H}D$ is the free Heyting algebra on the distributive lattice D. For the three-element chain $0 < p < 1$ the object Q is $(1 + 1)/E$ with the two points identified when $\neg p$; over the fork $\{1\} = \{1\} \subset \{p, 1\}$ of prime filters it has two points at 0 and one at 1, and no intuitionistically definable subobject of an object of $\mathsf{P}D$ behaves so.

6. Where the failure is: coproducts and quotients together

The pretopos completion is $\mathsf{P} = \mathsf{EF}$, with F the positive completion (finite disjoint coproducts) and E the exact completion of a regular category [1, A3.3], [4]. Here a coherent category is *positive* if it has finite coproducts, disjoint and stable under pullback, F is the reflector onto the positive coherent categories (coherent functors from $\mathsf{F}\mathcal{C}$ to a positive coherent category correspond to those from $\mathcal{C}$; the unit is full and faithful and full on subobjects, and every object of $\mathsf{F}\mathcal{C}$ is a finite coproduct of objects of $\mathcal{C}$), and E is the exact (ex/reg) completion of a regular category. The object Q uses a coproduct and a quotient. The coproduct step alone is never responsible;

the quotient step alone is not responsible for propositional theories, and is responsible under the hypothesis of Proposition 6.4.

Lemma 6.1. *If $\mathcal{D}$ is a Heyting category, then $\mathsf{F}\mathcal{D}$ is a Heyting category and the unit $\mathcal{D} \to \mathsf{F}\mathcal{D}$ is Heyting; and for a positive Heyting category $\mathcal{D}'$ a coherent functor $\mathsf{F}\mathcal{D} \to \mathcal{D}'$ is Heyting if and only if its restriction to $\mathcal{D}$ is.*

Proof. Every object of $\mathsf{F}\mathcal{D}$ is a finite coproduct $\coprod_i A_i$ of objects of $\mathcal{D}$, the unit is full and faithful and full on subobjects, and $\mathrm{Sub}(\coprod_i A_i) \cong \prod_i \mathrm{Sub}_{\mathcal{D}}(A_i)$, $U \mapsto (U \wedge A_i)_i$. Let $f\colon \coprod_i A_i \to \coprod_j B_j$, and let $A_{ij} = A_i \wedge f^* B_j$, a subobject of A_i in $\mathcal{D}$, and $f_{ij}\colon A_{ij} \to B_j$ the restriction, a morphism of $\mathcal{D}$; for each i the A_{ij} are disjoint with union A_i. For $V = (V_j)_j$ one has $f^*V \wedge A_i = \bigvee_j f_{ij}^* V_j$, and therefore, for $U \leq \coprod_i A_i$,

$$(\forall_f U) \wedge B_j \;=\; \bigwedge_i \forall_{f_{ij}} (U \wedge A_{ij}) :$$

$V \leq \forall_f U$ iff $f_{ij}^* V_j \leq U \wedge A_{ij}$ for all i, j iff $\bigvee_j f_{ij}^* V_j \leq U \wedge A_i$ for all i, the A_{ij} being disjoint. The formula involves only the $\forall$ of $\mathcal{D}$, finite meets and the decomposition of coproducts; so the unit preserves $\forall$, and so does a coherent functor to a positive Heyting category whose restriction to $\mathcal{D}$ is Heyting. The converse holds since the unit is Heyting. □

Proposition 6.2. *For every small coherent category $\mathcal{C}$ the canonical Heyting functor $\mathsf{HF}\mathcal{C} \to \mathsf{FH}\mathcal{C}$ is an equivalence.*

Proof. By Lemma 6.1, $\mathsf{FH}\mathcal{C}$ is a positive Heyting category with $\mathbf{Heyt}(\mathsf{FH}\mathcal{C}, \mathcal{D}') \simeq \mathbf{Heyt}(\mathsf{H}\mathcal{C}, \mathcal{D}') \simeq \mathbf{Coh}(\mathcal{C}, \mathcal{D}')$ for positive Heyting categories $\mathcal{D}'$. On the other side $\mathbf{Heyt}(\mathsf{HF}\mathcal{C}, \mathcal{D}') \simeq \mathbf{Coh}(\mathsf{F}\mathcal{C}, \mathcal{D}') \simeq \mathbf{Coh}(\mathcal{C}, \mathcal{D}')$. So it suffices to show that $\mathsf{HF}\mathcal{C}$ is positive; then both categories are the free positive Heyting category on $\mathcal{C}$, and the canonical functor, which is compatible with the functors from $\mathcal{C}$, is an equivalence. Let $X \leq \eta Y$ and $X' \leq \eta Y'$ be objects of $\mathsf{HF}\mathcal{C}$, with Y, Y' in $\mathsf{F}\mathcal{C}$. The coherent functor η sends the coproduct $Y + Y'$, the union of the disjoint subobjects Y and Y', to a union of disjoint subobjects, which in a coherent category is a coproduct, disjoint and stable [1, A1.4]; so $X \vee X' \leq \eta(Y + Y')$ is a disjoint and stable coproduct of X and X', and the initial object is the empty one. □

Corollary 6.3. *Let $\mathcal{C}$ be a small positive coherent category. Then $\mathsf{HE}\mathcal{C}$ is positive, and it is exact if and only if $\mathcal{C}$ is Boolean. For a small bounded distributive lattice D, on the other hand, $\mathsf{E}D \simeq D$ and $\mathsf{HE}D \simeq \mathsf{H}D$ is exact, although $\mathsf{HP}D$ is a pretopos only if D is Boolean.*

Proof. For positive $\mathcal{C}$ the category $\mathsf{E}\mathcal{C}$ is a pretopos, equivalent to $\mathsf{P}\mathcal{C}$ and to $\mathsf{FE}\mathcal{C}$; so $\mathsf{HE}\mathcal{C} \simeq \mathsf{HFE}\mathcal{C}$ is positive by the proof of Proposition 6.2, and a positive coherent category is a pretopos iff it is exact; Theorem 1.1 applies. In a preorder every equivalence relation on an object U is the whole of $U \times U \cong U$, the kernel pair of the identity; so D and the Heyting algebra $\mathsf{H}D$ are exact. □

For general coherent categories the quotient step alone does obstruct, as soon as a model at which a subobject is crossed has two points. For a coherent category $\mathcal{C}$ the exact completion $\mathsf{E}\mathcal{C}$ is coherent and effective (every equivalence relation is a kernel pair; a *cover* is a regular epimorphism), and restriction along the unit is an equivalence between the coherent functors from $\mathsf{E}\mathcal{C}$ and from $\mathcal{C}$ to an effective coherent category, with all natural transformations [4]; in particular $\mathrm{Mod}(\mathsf{E}\mathcal{C}) \simeq \mathrm{Mod}(\mathcal{C})$, and Lemma 3.1 holds for $\mathsf{HE}\mathcal{C}$ with the same proof: for an object X of $\mathsf{HE}\mathcal{C}$ and a Kripke model M of $\mathcal{C}$, the extension of M to $\mathsf{HE}\mathcal{C}$ sends X to a functor whose maps along the arrows w with $M(w)$ an isomorphism are injective.

Proposition 6.4. *Let $\mathcal{C}$ be a small coherent category, $S \leq A$ a subobject, $h\colon M \to N$ a homomorphism of models and $a \in A(M) \setminus S(M)$ with $h_A(a) \in S(N)$. If some object B of $\mathcal{C}$ has at least two points in M, then $\mathsf{HE}\mathcal{C}$ is not effective: the equivalence relation*

$$E\big((x,y),(x',y')\big) \;:=\; (x = x' \wedge y = y') \;\vee\; \big(x = x' \wedge \neg S(x)\big)$$

on $A \times B$ in $\mathsf{HE}\mathcal{C}$, which collapses the fibre over x when $\neg S(x)$, is not a kernel pair.

Proof. E is reflexive and symmetric, and transitive because each disjunct contains $x = x'$ and the second depends only on x. Suppose that E is the kernel pair of a morphism of $\mathsf{HE}\mathcal{C}$; replacing it by the cover in its image factorisation, E is the kernel pair of a cover $q\colon A \times B \to X$. Let M^* be the Kripke model over the fork of the proof of Theorem 1.1 and G its extension to $\mathsf{HE}\mathcal{C}$, a Heyting functor to $\mathbf{Set}^{\mathcal{K}}$. It preserves kernel pairs, covers and pseudo-complements, so $G(X)(k)$ is the quotient of $A(M^*_k) \times B(M^*_k)$ by the relation identifying (x,y) and (x,y') when $x \in (\neg M^*(S))(k)$. As in that proof, $a \notin (\neg M^*(S))(0)$ and $a \in (\neg M^*(S))(1)$; so for $y \neq y'$ in $B(M)$ the points (a,y) and (a,y') are distinct in $G(X)(0)$ and have the same image in $G(X)(1)$. This contradicts the invariant, $M^*(u)$ being an isomorphism. □

Example 6.5. For the theory of pure equality (Example 5.2) take $A = \sigma \times \sigma$, S the diagonal, the map from a two-element set to a one-element set, and $B = \sigma$: $\mathsf{HE}\mathcal{C}$ is not effective, so the Heyting completion does not commute with the exact completion, without any coproduct. In a bounded distributive lattice every object has at most one point in every model, and Proposition 6.4 does not apply, in accordance with Corollary 6.3.

So the Heyting completion commutes with adding coproducts; with adding quotients it commutes for propositional theories and does not commute when a subobject is crossed at a model in which some object has two points. Whether it commutes with adding quotients in the remaining cases, those of the non-Boolean categories in which every model at which a subobject is crossed gives at most one point to every object, is open (Question 8.1).

7. When excluded middle repairs the comparison

Let $\mathsf{K}\colon \mathbf{Heyt} \to \mathbf{BCoh}$ be the classicalisation, the reflector onto the full sub-2-category $\mathbf{BCoh}$ of Boolean coherent categories which adds excluded middle to the internal first-order theory; $\mathsf{B} \simeq \mathsf{KH}$ is the Morleyization, and $\mathsf{KP}\mathcal{D} \simeq \mathsf{PK}\mathcal{D}$ for every Heyting category $\mathcal{D}$ [3, Thm. 9.2]. The comparison $b_{\mathcal{C}}\colon \mathsf{BP}\mathcal{C} \to \mathsf{PB}\mathcal{C}$ of [3] is $\mathsf{K}c_{\mathcal{C}}$ followed by this equivalence at $\mathcal{D} = \mathsf{H}\mathcal{C}$ [3, Cor. 9.3]:

$$\begin{array}{ccccc} \mathsf{HP}\mathcal{C} & & \xrightarrow{\quad c_{\mathcal{C}} \quad} & & \mathsf{PH}\mathcal{C} \\ \downarrow & & & & \downarrow \\ \mathsf{BP}\mathcal{C} = \mathsf{KHP}\mathcal{C} & & \xrightarrow{\mathsf{K}c_{\mathcal{C}}} & \mathsf{KPH}\mathcal{C} & \xrightarrow{\quad \simeq \quad} \mathsf{PKH}\mathcal{C} = \mathsf{PB}\mathcal{C}. \end{array}$$

$$\mathsf{BP}\mathcal{C} \xrightarrow{\quad b_{\mathcal{C}} \quad} \mathsf{PB}\mathcal{C}$$

Corollary 7.1. *For a small coherent category $\mathcal{C}$ the following are equivalent: the Morleyization commutes with the pretopos completion at $\mathcal{C}$, that is $b_{\mathcal{C}}$ is an equivalence; the Galois criterion of* [3, Thm. 6.1] *holds for $\mathcal{C}$ (in the saturated, strongly homogeneous models of the completions of the classical theory, every classical imaginary has the automorphism stabiliser of a point of a coherent imaginary); the Heyting functor $c_{\mathcal{C}}$ is inverted by K. If these conditions hold and $\mathcal{C}$ is not Boolean, then $c_{\mathcal{C}}$ is a full and faithful Heyting functor which is not an equivalence and becomes one when excluded middle is added. This is the case for the theory of pure equality, for the theory of an equivalence relation, for the theories without axioms over a monadic relational signature with at least one sort, and for the non-Boolean propositional theories* [3, Ex. 6.5, Thm. 8.2, Thm. 7.1, Prop. 5.2].

Proof. The first equivalence is [3, Thm. 6.1], the second the factorisation above, and the rest is Theorem 1.1. The theories listed are not Boolean: in each there is a homomorphism of models which is not an elementary embedding (a non-injective map of sets in the first and third cases, a map merging two classes, a proper inclusion of prime filters). □

In the example of pure equality one sees the repair: classically $\neg(x = y)$ is a complemented subobject of $\sigma \times \sigma$ and the glued object Q of Example 5.2 becomes the coproduct of $\sigma \times \sigma$ and the diagonal, a classical subobject of a coherent imaginary. For the theory of complete multipartite graphs [3, Thm. 5.3] the repair fails: K does not invert $c_{\mathcal{C}}$. Thus the statement of [3] that the forced order of the Morleyization and the completion arises in the Heyting completion is to be read as follows. The comparison is never an equivalence before excluded middle is added, except trivially; the Galois criterion decides whether excluded middle repairs it.

8. Questions

Question 8.1. Let $\mathcal{C}$ be a non-Boolean coherent category such that, whenever a homomorphism $h\colon M \to N$ crosses a subobject, every object of $\mathcal{C}$ has

at most one point in M (for instance the product of a non-Boolean bounded distributive lattice and a Boolean coherent category). Is $\mathsf{HE}\mathcal{C}$ effective?

Question 8.2. Describe the essential image of $c_{\mathcal{C}}$ in $\mathsf{PH}\mathcal{C}$. The invariant of Lemma 3.1 is necessary; is it sufficient, for instance for propositional theories, where $\mathsf{PH}D$ is the pretopos completion of the free Heyting algebra on D?

Question 8.3. Let L be an intermediate predicate logic, given by axiom schemes added to intuitionistic first-order logic, and H_L the corresponding translation. The fork is not a frame for the logic of linear frames (Gödel–Dummett), and over a finite linear order two consecutive nodes whose transition is an isomorphism of models force the same formulas, under this isomorphism; so the proof of Theorem 1.1 does not apply. For which L does H_L commute with P beyond the Boolean case?

Question 8.4. Is there a description of the class of coherent functors inverted by H, and of its relation to the classes V_{B} and V_{P} of [3] in the lattice of two-out-of-three classes, in spite of H not being a reflector?

Declarations

The author has no competing interests to declare.

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